Experiment 9

Travelling Salesman Problem (TSP) – Python Program

Aim

To write a Python program to find the minimum cost path using the **Travelling Salesman Problem (TSP)**.

Algorithm

1. Start.
2. Read the number of cities.
3. Read the cost matrix.
4. Generate all possible routes starting from city 0.
5. Calculate the total cost for each route.
6. Find the route with the minimum cost.
7. Display the minimum cost.
8. Stop.

Program

```
from itertools import permutations


n = int(input("Enter number of cities: "))

cost = []


print("Enter cost matrix:")

for i in range(n):

  cost.append(list(map(int, input().split())))


cities = list(range(1, n))

min_cost = float('inf')


for path in permutations(cities):
```

```
    total = 0
    k = 0
    for city in path:
        total += cost[k][city]
        k = city
    total += cost[k][0]
    min_cost = min(min_cost, total)

print("Minimum Cost:", min_cost)
```

Input

Enter number of cities: 4

Enter cost matrix:

0 10 15 20

10 0 35 25

15 35 0 30

20 25 30 0

Output

```
Enter number of cities: 4
Enter cost matrix:
0 10  15 20
10 0 35 25
15 35 0 30
20 25 30 0
Minimum Cost: 80
```

Result

Thus, the Python program to implement the Travelling Salesman Problem (TSP) was executed successfully, and the minimum travelling cost was obtained.